DGMS 2019 GEOLOGY • QUESTION 44

Crystallography: Hermann-Mauguin Symbols & Symmetry Axis Matching

Topic: Geometric Crystallography & Point Groups | Primary Reference: Dexter Perkins (Mineralogy, 3rd ed., Ch. 9 & 10)

Question Statement

Q.44. Match Hermann-Mauguin notations List-I with List-II (Crystal class):

List-I (Symmetry Axes / Elements):
A. $3A_4$, $4A_3$, $6A_2$
B. $1A_4$, $4A_2$
C. $1A_2$, 2m
D. $1A_6$, $6A_2$

List-II (Hermann-Mauguin Class Symbol):
1. 4 2 2
2. 4 3 2
3. m m 2
4. 6 2 2

(a) A-2, B-1, C-3, D-4
(b) A-1, B-2, C-4, D-3
(c) A-4, B-3, C-2, D-1
(d) A-3, B-4, C-1, D-2

a **A-2, B-1, C-3, D-4 (Correct Answer)**

b A-1, B-2, C-4, D-3

c A-4, B-3, C-2, D-1

d A-3, B-4, C-1, D-2

✓ **Correct Answer: (a) A-2, B-1, C-3, D-4 [$3A_4$, $4A_3$, $6A_2$ = 4 3 2 (Cubic); $1A_4$, $4A_2$ = 4 2 2 (Tetragonal); $1A_2$, 2m = m m 2 (Orthorhombic); $1A_6$, $6A_2$ = 6 2 2 (Hexagonal)].**

Textbook Evidence: Dexter Perkins (Chapter 9 & 10)

BOX 7

Hermann-Mauguin Symbols

The symbols used in this text are based on notations developed by C. H. Hermann and C. V. Mauguin in the early 1900s. They have been used by most crystallographers since about 1930. Numbers refer to rotation axes of symmetry; a bar over a number indicates a rotoinversion axis. Mirrors, designated by *m*, are perpendicular to an axis if they appear as a denominator (for example,4/*m*), and parallel to an axis otherwise. When articulating the symbols, they are pronounced just as if they were typographical characters. 4/*m*$\bar{3}$2/*m,* for example, is read "four over m, bar three, two over m."

One, two, or three symbols describe a point group; they combine in different ways for different systems (Figure 34):

For cubic point groups, the first symbol describes three mutually perpendicular principal symmetry axes, oriented perpendicular to cubic faces (if cubic faces are present). The second describes four axes oriented at angles of 54°44′ to the principal axes. They correspond to the "body diagonals" of a cube, a diagonal from a corner through the center to the opposite corner. The third symbol, if present, describes six 2-fold axes or mirror planes oriented at angles of 45° to the principal axes. They correspond to "edge diagonals" of a cube, diagonals from the center of edges through the center of the cube to the opposite edge.

For hexagonal point groups, the first symbol describes the single principal axis. The second, if present, describes three secondary rotation axes oriented at 120° to each other and perpendicular to the principal axis, or three mirror planes oriented at 120° to each other and parallel to the principal axis. The third symbol, if present, represents mirror planes or 2-fold axes oriented between the secondary axes.

For tetragonal point groups, the first symbol represents the principal axis. The second, if present, represents two secondary axes perpendicular to each other and to the principal axis, or two mirror planes oriented at 90° to each other and parallel to the principal axis. The third represents axes or mirror planes between the secondary axes.

Only three orthorhombic point groups are possible. Point group 222 has three mutually perpendicular 2-fold axes. Point group *mm*2 has one 2-fold axis with two mutually perpendicular mirror planes parallel to it. Point group 2/*m*2/*m*2/*m* has three perpendicular 2-fold axes with mirror planes perpendicular to each.

For monoclinic point groups, only one symmetry element is included in the Hermann-Mauguin symbols because the only possible symmetries are a 2-fold axis, a mirror, or a 2-fold axis with a mirror perpendicular to it. Similarly, for triclinic crystals, the only possible point groups are 1 and $\bar{1}$.

Crystal System	Possible First Symbol	Possible Second Symbol	Possible Third Symbol
cubic	4, 4/*m*, $\bar{4}$,2, 2/*m*	3,$\bar{3}$,	2, 2/*m*, m
hexagonal	6, 6/*m*,$\bar{6}$, 3, $\bar{3}$	2,2/*m*, *m*	2, 2/*m*, *m*
tetragonal	4, 4/*m*, $\bar{4}$,	2,2/*m*, *m*	2, 2/*m*, *m*
orthorhombic	2, 2/*m*, *m*	2,2/*m*, *m*	2/*m*, *m*
monoclinic	2, 2/*m*, *m*		
triclinic	1, $\bar{1}$		

TABLE 3 The Possible Forms of Each Point Group

Triclinic-Monoclinic-Orthorhombic Systems

Box 9.7: International Hermann-Mauguin Crystallographic Symbols (Dexter Perkins, *Mineralogy*, 3rd ed., p. 276). Rules governing point group symbols: Numbers ($2, 3, 4, 6$) denote proper rotation axes; m denotes mirror planes. For **cubic** crystals, the second symbol represents the four 3-fold axes along the body diagonals (e.g., 432). For **tetragonal**, the first position is the principal 4-fold axis along c (422). For **hexagonal**, the first position is the principal 6-fold axis along c (622). For **orthorhombic**, the three positions correspond to axes along a, b, c ($mm2$).

> *"In the Hermann-Mauguin system:*
> • **Cubic (4 3 2):** *Three 4-fold axes parallel to a, b, c, four 3-fold axes along cube body diagonals, and six 2-fold axes along edge diagonals.*
> • **Tetragonal (4 2 2):** *One 4-fold principal axis along c, two 2-fold axes along a_1, a_2, and two 2-fold diagonal axes.*
> • **Orthorhombic (m m 2):** *Two mutually perpendicular vertical mirror planes intersecting along a single 2-fold rotation axis.*
> • **Hexagonal (6 2 2):** *One 6-fold principal axis along c, three 2-fold axes along a_1, a_2, a_3, and three intermediate 2-fold axes."*
>
> **— Dexter Perkins, *Mineralogy*, Chapter 9 & 10, pp. 270–276**

Detailed Step-by-Step Point Group Matching

A. $3A_4$, $4A_3$, $6A_2$ → 2. 4 3 2 (Isometric / Cubic Gyroidal Class) [MATCH 1]

Symmetry Elements: Three 4-fold axes of rotation ($3A_4$), four 3-fold axes of rotation ($4A_3$), six 2-fold axes of rotation ($6A_2$). Center of symmetry and mirror planes are absent (C absent, P absent). In the Hermann-Mauguin system, the 4-fold axes along a, b, c form the 1st position (4), the 3-fold body diagonal axes form the 2nd position (3), and the 2-fold edge diagonal axes form the 3rd position (2), giving **4 3 2**.

B. $1A_4$, $4A_2$ → 1. 4 2 2 (Tetragonal Trapezohedral Class) [MATCH 2]

Symmetry Elements: One 4-fold principal axis along the c-crystallographic axis ($1A_4$), four 2-fold axes perpendicular to c ($4A_2$: two along a_1, a_2 and two along the bisecting diagonal directions). No mirror planes (m) and no center of symmetry. The Hermann-Mauguin notation is **4 2 2**.

C. $1A_2$, $2m$ → 3. m m 2 (Orthorhombic Pyramidal Class) [MATCH 3]

Symmetry Elements: One 2-fold rotation axis ($1A_2$, conventionally taken along c) and two vertical mirror planes ($2m$) intersecting along the 2-fold axis ($m \perp a$ and $m \perp b$). No horizontal mirror plane and no center of symmetry. Written as **m m 2** (or $mm2$).

D. $1A_6$, $6A_2$ → 4. 6 2 2 (Hexagonal Trapezohedral Class) **[MATCH 4]**

Symmetry Elements: One 6-fold principal rotation axis along c ($1A_6$), six 2-fold axes perpendicular to c ($6A_2$: three along a_1, a_2, a_3 and three intermediate diagonal axes). No mirror planes and no inversion center. Represented in Hermann-Mauguin notation as **6 2 2**.

Hermann-Mauguin Position Rules Across Crystal Systems

Crystal System	1st Position in Symbol	2nd Position in Symbol	3rd Position in Symbol	Example Class (from Q.44)
Isometric (Cubic)	3 mutually perpendicular cube axes (a_1, a_2, a_3)	**4 body diagonals (3-fold axes)**	6 face / edge diagonals	**4 3 2 ($3A_4$, $4A_3$, $6A_2$)**
Tetragonal	**Principal c-axis (4-fold)**	2 axial axes (a_1, a_2)	2 diagonal axes	**4 2 2 ($1A_4$, $4A_2$)**
Hexagonal	**Principal c-axis (6-fold)**	3 coplanar axes (a_1, a_2, a_3)	3 intermediate diagonal axes	**6 2 2 ($1A_6$, $6A_2$)**
Orthorhombic	Symmetry along a-axis	Symmetry along b-axis	Symmetry along c-axis	**m m 2 ($1A_2$, 2m)**

Key Takeaway & Exam Strategy

Universal Hermann-Mauguin Recognition Shortcuts:

- Presence of a **3 in the second position** (e.g., $432, m\bar{3}m, 23$) $\rightarrow$ Always **Cubic (Isometric)** system!
- Starting with a **4** (e.g., $422, 4/m, 4mm$) $\rightarrow$ Always **Tetragonal** system!
- Starting with a **6** (e.g., $622, 6/m, 6mm$) $\rightarrow$ Always **Hexagonal** system!

- Symbols composed exclusively of 2s and ms (e.g., $222, mm2, 2/m 2/m 2/m$) $\rightarrow$ Always **Orthorhombic** system!
- Matching: **A-2, B-1, C-3, D-4** (Option a).